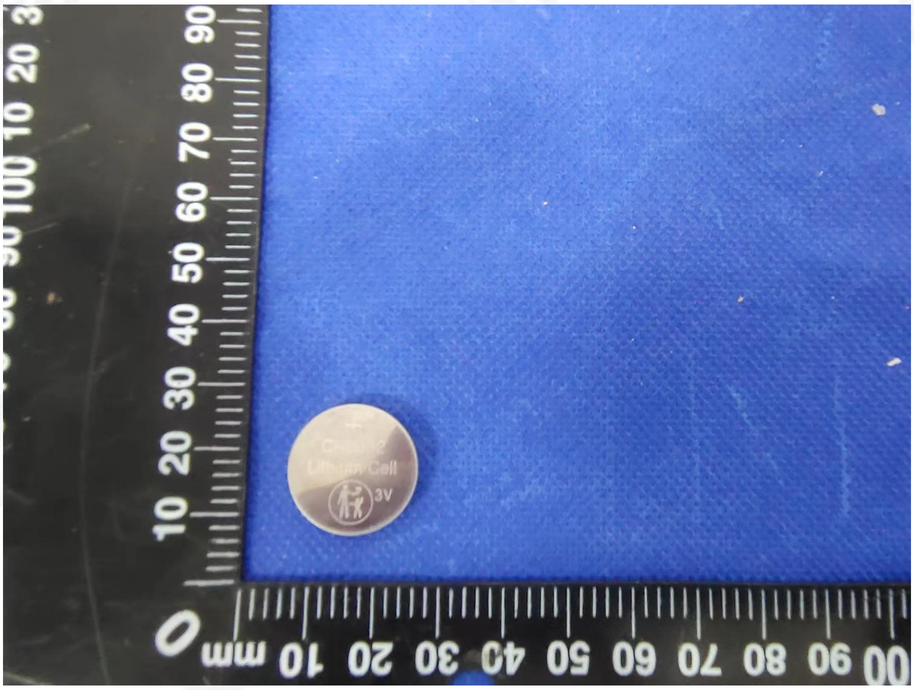
TEST REPORT
IEC/EN 60086-1 & IEC/EN 60086-2
Primary Battery - Part 1: General
Primary Battery - Part 2: Physical and electrical specifications

Report Number:	ZKT-2204122353S
Date of issue:	Apr. 19, 2022
Total number of pages:	21
Name of Testing Laboratory preparing the Report:	**Shenzhen ZKT Technology Co., Ltd.**
Address:	1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China
Applicant's name:	**Luoyang Anxin New Energy Technology Co. LTD**
Address:	Anxin Industrial Park, Dongdajiao Village, Luolong District, Luoyang city, Henan Province
Test specification:	
Standard:	IEC 60086-1:2015 in conjunction with IEC 60086-2:2015 EN 60086-1:2015 in conjunction with EN 60086-2:2016
Non-standard test method:	N/A
Test Report Form No.:	TRF-EL-059_V0
Test Report Form(s) Originator:	ZKT Testing
Master TRF:	Dated: 2020-01-06
Test item description:	Lithium manganese buckle battery CR2032
Trade Mark:	ETEnergy
Manufacturer:	Same as applicant
Model/Type reference:	CR2032
Ratings:	3V, 220mAh
Designation:	Category 4 batteries (CR2032)

Testing procedure and testing location:

Testing Laboratory......................................:	**Shenzhen ZKT Technology Co., Ltd.**
Address...:	1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China
Date of Test...:	Apr. 12, 2022 to Apr. 19, 2022
Tested by (name + signature)................:	Peter Huang Peter Huang
Reviewed by (name + signature)..........:	Simon Gong Simon Gong
Approved by(name + signature)...........:	Awen He Awen Shenzhen ZKT Technology Co., Ltd. Approved

Copy of marking plate (representative)
The artwork below may be only a draft.



Test item particulars... :	
Classification of installation and use.......................:	Built-in and use in the fit battery compartment
Supply Connection...:	Positive negative connection
Possible test case verdicts:	
- test case does not apply to the test object...........:	N/A
- test object does meet the requirement..................:	P (Pass)
- test object does not meet the requirement...........:	F (Fail)
Testing.. :	
Date of receipt of test item..:	2022-04-12
Date (s) of performance of tests...............................:	2022-04-12 to 2021-04-19

General remarks:

"(See Enclosure #)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

Manufacturer's Declaration per sub-clause 4.2.5 of IECEE 02:

The application for obtaining a CB Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided..:	☐ **Yes** ☒ **Not applicable**

When differences exist; they shall be identified in the General product information section.

Name and address of factory (ies)...........................:	Camelion Battery Co., Ltd. Unit 705, Cyber Times Tower A, Tian'an Cyber Park, Shenzhen, China. 518041

General product information:

(1) The equipment under test (EUT) is a LITHIUM MANGANESE BUTTON CELL.
(2) The battery Standardized chemical system letter is C, Chemical System is Lithium-Manganese Dioxide (Organic electrolyte), mercury and cadmium and lead are not added.
(3) IEC designation is CR2032, common designation is 2032.
(4) Dimension of the battery: (ф) 16mm by (A) 3.2mm max.
(5) The battery weight approx. 2.9g.

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
4	**REQUIREMENTS**		**P**
4.1	**General**	See below.	**P**
4.1.1	Design	Complied.	P
	Additionally, information on equipment design can be found in Annex B	It should evaluation in the equipment designers.	N/A
4.1.2	Battery dimensions	Complied.	P
	The dimensions for individual types of batteries are given in IEC 60086-2 and IEC 60086-3	Complied. (See table 6.4.6)	P
4.1.3	Terminals	Considered.	P
4.1.3.1	General	See below.	P
	Terminals shall be in accordance with Clause 6 of IEC 60086-2	Complied. (See table 6.4.6)	P
	Their physical shape shall be designed in such a way that they ensure that the batteries make and maintain good electrical contact at all times	Complied.	P
	They shall be made of materials that provide adequate electrical conductivity and corrosion protection	Complied.	P
4.1.3.2	Contact pressure resistance	Complied.	P
	Where stated in the battery specification tables or the individual specification sheets in IEC 60086-2, the following applies		P
	A force of 10 N applied through a steel ball of 1 mm diameter at the centre of each contact area for a period of 10 s shall not cause any apparent deformation which might prevent satisfactory operation of the battery		P
4.1.3.3	Cap and base	Category 4- Specifications: CR2032. Not such Category.	N/A
	This type of terminal is used for batteries which have their dimensions specified according to Figures 1 and 7 of IEC 60086-2 and which have the cylindrical side of the battery insulated from the terminals	The cylindrical surface is connected to the positive terminal.	N/A
4.1.3.4	Cap and case	Complied.	P
	This type of terminal is used for batteries which have their dimensions specified according to Figures 8, 9, 10, 14, 15 and 16 of IEC 60086-2, but in which the cylindrical side of the battery forms part of the positive terminal	The cylindrical surface is connected to the positive terminal.	P
4.1.3.5	Screw terminals	Category 4- Specifications: CR2032. Not such Category.	N/A
	This contact consists of a threaded rod in combination with either a metal or insulated metal nut		N/A
4.1.3.6	Flat contacts	Complied.	P

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	These are essentially flat metal surfaces adapted to make electrical contact by suitable contact mechanisms bearing against them		P
4.1.3.7	Flat or spiral springs	Category 4- Specifications: CR2032. Not such Category.	N/A
	These contacts comprise flat metal strips or spirally wound wires which are in a form that provides pressure contact		N/A
4.1.3.8	Plug-in-sockets	Category 4- Specifications: CR2032. Not such Category.	N/A
	These are made up of a suitable assembly of metal contacts, mounted in an insulated housing or holding device and adapted to receive corresponding pins of a mating plug		N/A
4.1.3.9	Snap fasteners	Category 4- Specifications: CR2032. Not such Category.	N/A
4.1.3.9.1	General	Category 4- Specifications: CR2032. Not such Category.	N/A
	These contacts are composed of a combination comprising a stud (non-resilient) for the positive terminal and a socket (resilient) for the negative terminal		N/A
	They shall be of suitable metal so as to provide efficient electrical connection when joined to the corresponding parts of an external circuit		N/A
4.1.3.9.2	Snap fastener	Category 4- Specifications: CR2032. Not such Category.	N/A
	This type of terminal consists of a stud for the positive terminal and a socket for the negative terminal. These shall be made from nickel plated steel or other suitable material		N/A
	They shall be designed to provide a secure physical and electrical connection, when fitted with similar corresponding parts for connection to an electrical circuit		N/A
4.1.3.10	Wire	Not wire used. It should evaluation by equipment designers.	N/A
	Wire leads shall be single or multi-strand flexible insulated tinned copper		N/A
	The positive terminal wire covering shall be red and the negative black		N/A
4.1.3.11	Other spring contacts or clips	Not Other spring contacts or clips used. It should evaluation by equipment designers.	N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	These contacts are generally used on batteries when the corresponding parts of the external circuit are not precisely known		N/A
	They shall be of spring brass or of other material having similar properties		N/A
4.1.4	Classification (electrochemical system)	Complied.	P
	Primary batteries are classified according to their electrochemical system	Standardized electrochemical systems: “C”.	P
	Each system, with the exception of the zinc-ammonium chloride, zinc chloride-manganese dioxide system, has been allocated a letter denoting the particular system	Category 4- Specifications: CR2032	P
	The electrochemical systems that have been standardized up to now are given in Table 1	Category 4- Specifications: CR2032	P
4.1.5	Designation	Complied.	P
	The designation of primary batteries is based on their physical parameters, their electrochemical system as well as modifiers, if needed	Complied.	P
	A comprehensive explanation of the designation system (nomenclature) can be found in Annex C	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.1.6	Marking	Complied.	P
4.1.6.1	General (see Table 2)	Complied.	P
	With the exception of small batteries (see 4.1.6.2), each battery shall be marked with the following information	Complied. See copy of marking plate.	P
	a) designation, IEC or common.................................:	IEC: CR2032. Common: 1632	P
	b) expiration of a recommended usage period or year and month or week of manufacture..................:	Small battery	N/A
	The year and month or week of manufacture may be in code... :	Small battery	N/A
	c) polarity of the positive (+) terminal........................:	“+”, “-” Marked on the battery.	P
	d) nominal voltage..:	3V Marked on the battery.	P
	e) name or trade mark of the manufacturer or supplier... :	Camelion Marked on the battery.	P
	f) cautionary advice..:	Small battery	N/A
4.1.6.2	Marking of small batteries (see Table 2)	Complied.	P
	a) Batteries designated in IEC as small, mainly category 3 and category 4 batteries have a surface too small to accommodate all markings shown in 4.1.6.1	Complied.	P

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict

	For these batteries the designation 4.1.6.1a) and the polarity 4.1.6.1c) shall be marked on the battery.... :	IEC: CR2032. Common: 1632 “+”, “-” Marked on the battery.	P
	All other markings shown in 4.1.6.1 may be given on the immediate packing instead of on the battery.....:		P
	b) For P-system batteries, 4.1.6.1a) may be on the battery, the sealing tab or the package................... :	Not P-system batteries	N/A
	4.1.6.1c) may be marked on the sealing tab and/or on the battery...:		P
	4.1.6.1b), 4.1.6.1d) and 4.1.6.1e) may be given on the immediate packing instead of on the battery.....:		P
	c) Caution for ingestion of swallowable batteries shall be given..:		P
4.1.6.3	Marking of batteries regarding method of disposal	Manufacturer declarate it was provided to equipment designers which will show in the equipment use manual.	P
	Marking of batteries with respect to the method of disposal shall be in accordance with local legal requirements	Complied.	P
4.1.7	Interchangeability: battery voltage	Complied.	P
	Voltage range 1, Ur = 1,4 V: Batteries having a standard discharge voltage m × Us equal to or within the range of n × 1,19 V to n × 1,61 V.............:	Not this Voltage range.	N/A
	Voltage range 2, Ur = 3,2 V: Batteries having a standard discharge voltage m × Us equal to or within the range of n × 2,72 V to n × 3,68 V.......................:	"C" system. standard discharge voltage Us: 2.9V.	P
4.2	**Performance**	See below.	**P**
4.2.1	Discharge performance	Complied.	P
	Discharge performance of primary batteries is specified in IEC 60086-2	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.2.2	Dimensional stability	Complied.	P
	The dimensions of batteries shall conform with the relevant specified dimensions as given in IEC 60086-2 and IEC 60086-3 at all times during discharge testing under the standard conditions given in this specification	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.2.3	Leakage	See below.	P
	When batteries are stored and discharged under the standard conditions given in this specification, no leakage shall occur	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.2.4	Open-circuit voltage limits	See below.	P

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The maximum open-circuit voltage of batteries shall not exceed the values given in Table 1.................... :	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.2.5	Service output	See below.	P
	Discharge durations, initial and delayed, of batteries shall meet the requirements given in IEC 60086-2	Category 4- Specifications: CR2032 (See table 6.4.6)	P
4.2.6	Safety	Complied.	P
	When designing primary batteries, safety under conditions of intended use and foreseeable mis-use as prescribed in IEC 60086-4 and IEC 60086-5 shall be considered	IEC 60086-4 was evaluated in the ZKT-2204122506S issued by Shenzhen ZKT Technology Co., Ltd.	P
5	**PERFORMANCE – TESTING**		**P**
5.1	**General**	Category 4- Specifications: CR2032 (See table 6.4.6)	**P**
	For the preparation of standard methods of measuring performance (SMMP) of consumer goods, refer to Annex E	Complied.	P
	A capacity of a primary battery may be established by electrical discharge tests as detailed in D.2.3	Complied.	P
	However, under consumer usage conditions, the capacities realised from electrical discharge test methods can vary	Complied.	P
	The following factors/variables dramatically impact on optimum capacity realisation	Complied.	P
	a) The current demand from the external electrical circuit/device	Complied.	P
	b) The frequency of current demand. (Continuous or intermittent usage)	Complied.	P
	c) The minimum voltage at which the device will satisfactorily operate. (Cut off voltage)	Complied.	P
	d) The temperature of operation	Complied.	P
5.2	**Discharge testing**	Complied. (See table 6.4.6)	**P**
5.2.1	General	See below.	P
	The discharge tests in this standard fall into two categories	See below.	P
	– application tests	Service output tests.	N/A
	– service output tests	Complied. (See table 6.4.6)	P
5.2.2	Application tests	Service output tests.	N/A
5.2.2.1	General		N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Application tests may be accelerated by discharge load, daily period duty cycle, or both. The specified values for load and time intermittency should take the following factors into consideration		N/A
	– discharge efficiency of the battery relative to the application		N/A
	– typical duty cycle use patterns for the application		N/A
	– total time to conduct the test not to exceed 30 days		N/A
5.2.2.2	Application tests with multiple loads		N/A
	For application test with multiple loads, the load order during a cycle shall start with the heaviest load and move to the lightest load unless otherwise specified		N/A
5.2.3	Service output tests		N/A
	For service output tests, the value of the load resistor should be selected such that the service output approximates 30 days		N/A
5.3	**Conformance check to a specified minimum average duration**	Complied.	**P**
	In order to check the conformance of a battery to any discharge tests specified in IEC 60086-2 and 60086-3, the test shall be carried out as follows	Complied. (See table 6.4.6)	P
	a) Test eight batteries	Complied.	P
	b) Calculate the average without the exclusion of any result	Complied.	P
	c) If this average is equal to or greater than the specified figure and no more than one battery has a service output of less than 80 % of the specified figure, the batteries are considered to conform to service output		N/A
	d) If this average is less than the specified figure and/or more than one battery has a service output of less than 80 % of the specified figure, repeat the test on another sample of eight batteries and calculate the average as previously		N/A
	e) If the average of this second test is equal to or greater than the specified figure and no more than one battery has a service output of less than 80 % of the specified figure, the batteries are considered to conform to service output		N/A
	f) If the average of the second test is less than the specified figure and/or more than one battery has a service output of less than 80 % of the specified figure, the batteries are considered not to conform and no further testing is permitted		N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	g) For the purposes of verifying compliance with this standard, conditional acceptance may be given after completion of the initial discharge tests		N/A
5.4	**Calculation method of the specified value of a minimum average duration**	Complied. (See table 6.4.6)	**P**
	This method is described in Annex F	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	P
5.5	**OCV testing**	Complied. (See table 6.4.6)	**P**
	Open-circuit voltage shall be measured with the voltage measuring equipment specified in 6.8.1	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	P
5.6	**Battery dimensions**	Complied. (See table 6.4.6)	**P**
	Dimensions shall be measured with the measuring equipment specified in 6.8.2	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	P
5.7	**Leakage and deformation**	Complied. (See table 6.4.6)	**P**
	After the service output has been determined under the specified environmental conditions, the discharge shall be continued in the same way until the closed circuit voltage drops for the first time below 40 % of the nominal voltage of the battery	Complied. (See table 6.4.6)	P
	The requirements of 4.1.3, 4.2.2 and 4.2.3 shall be met	Complied. (See table 6.4.6)	P
6	**PERFORMANCE – TEST CONDITIONS**	Complied.	**P**
6.1	**Storage and discharge conditions**	See below.	**P**
	Storage before discharge testing and the actual discharge test is carried out under well-defined conditions. Unless otherwise specified, the conditions given in Table 3 shall apply	Complied. (See table 6.4.6)	P
	Initial discharge test, storage condition Temperature: 20±2 °C Relative humidity: 55±20 % RH................................:	(See appended table)	P
	Initial discharge test, discharge condition Temperature: 20±2 °C Relative humidity: 55+20/-40 % RH.........................:	(See appended table)	P
	Delayed discharge test, storage condition Temperature: 20±2 °C Relative humidity: 55±20 % RH................................:	(See appended table)	N/A
	Delayed discharge test, discharge condition Temperature: 20±2 °C Relative humidity: 55+20/-40 % RH.........................:	(See appended table)	N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Delayed discharge test (high temperature), storage condition Temperature: 45±2 °C Relative humidity: 55±20 % RH................................:	(See appended table)	N/A
	Delayed discharge test (high temperature), discharge condition Temperature: 20±2 °C Relative humidity: 55+20/-40 % RH.........................:	(See appended table)	N/A
6.2	**Commencement of discharge tests after storage**	See below.	**P**
	The period between the completion of storage and the start of a delayed discharge test shall not exceed 14 days	Complied	P
	During this period the batteries shall be kept at (20 ± 2) °C and (55 + 20 / -40) % RH (except for P-system batteries where the relative humidity shall be (55 ± 10) % RH)	Complied	P
	At least one day in these conditions shall be allowed for normalization before starting a discharge test after storage at high temperature	Complied. (See table 6.4.6)	P
6.3	**Discharge test conditions**	Complied. (See table 6.4.6)	**P**
6.3.1	General	See below.	P
	In order to test a battery it shall be discharged as specified in IEC 60086-2 or IEC 60086-3 until the voltage on load drops for the first time below the specified end-point	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	P
6.3.2	Compliance	Complied. (See table 6.4.6)	P
	When IEC 60086-2 or IEC 60086-3 specify service outputs for more than one discharge test, batteries shall meet all of these requirements in order to comply with this specification	Four discharge test complied the standard requirements.	P
6.4	**Load resistance**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	When formulating new tests, the resistive load shall, whenever possible, be as shown in Table 4 together with their decimal multiples or sub-multiples	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	N/A
6.5	**Time periods**	Complied.	**P**
	The periods on-discharge and off-discharge shall be as specified in IEC 60086-2	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	P
	When formulating new tests, whenever possible, one of the following daily periods should be adopted from Table 5...:		N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Other cases are specified in IEC 60086-2, if necessary		N/A
6.6	**Test condition tolerances**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	Unless otherwise specified, the tolerances given in Table 6 shall apply	Complied.	P
6.7	**Activation of ‘P’-system batteries**	“C” system batteries.	**N/A**
	A period of at least 10 min shall elapse between activation and the commencement of electrical measurement		N/A
6.8	**Measuring equipment**	Complied.	**P**
6.8.1	Voltage measurement	Complied.	P
6.8.2	Mechanical measurement	Complied.	P
7	**SAMPLING AND QUALITY ASSURANCE**	See below.	**P**
	The use of sampling plans or product quality indices should be agreed between the manufacturer and the purchaser	Complied.	P
	Where no agreement is specified, refer to ISO 2859 and ISO 21747 for sampling and quality compliance assessment advice	The use of sampling plans or product quality indices should be agreed between the manufacturer and the purchaser.	N/A
8	**BATTERY PACKAGING**	Considered.	**P**
	A code of practice for battery packaging, shipment, storage, use and disposal can be found in Annex G	Complied.	P
ANNEX A	**CRITERIA FOR THE STANDARDIZATION OF BATTERIES**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	Batteries and electrochemical systems shall meet the following requirements to justify their initial inclusion or ongoing retention in the IEC 60086 series		P
	a) The battery or batteries of this electrochemical system are in mass production		P
	b) The battery or batteries of this electrochemical system are available in several market places of the world		P
	c) The battery is produced by at least two independent manufacturers, the patent holder(s) of which shall conform to the requirements contained in 2.14 of the ISO/IEC Directives, Part 1, Reference to patented items		P

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	d) The battery is produced in at least two different countries or, alternatively, the battery is purchased by other international and independent battery manufacturers and sold under their company label		P
	The items of Table A.1 shall be included in any new work proposal to standardize a new individual battery or electrochemical system		P
ANNEX B	**RECOMMENDATIONS FOR EQUIPMENT DESIGN**	Provided in the battery specification, It should be considered by battery-powered equipment designer.	**P**
B.1	**Technical liaison**		**N/A**
	It is recommended that companies producing battery-powered equipment maintain close liaison with the battery industry. The capabilities of existing batteries should be taken into account at design inception. Whenever possible, the battery type selected should be one included in IEC 60086-2		N/A
	The equipment should be permanently marked with the IEC designation, grade and size of battery which will give optimum performance		N/A
B.2	**Battery compartment**		**N/A**
B.2.1	General		N/A
	Design compartments so that batteries are easily inserted and do not fall out. The dimensions and design of compartments and contacts should be such that batteries complying with this standard will be accepted		N/A
	The design of the negative contact should make allowance for any recess of the battery terminal		N/A
	Clearly indicate the type of battery to use, the correct polarity alignment and directions for insertion		N/A
	Use the shape and/or the dimensions of the positive (+) and negative (–) battery terminals in compartment designs to prevent the reverse connection of batteries		N/A
	Positive (+) and negative (–) battery contacts should be visibly different in form to avoid confusion when inserting batteries		N/A
	Battery compartments should be electrically insulated from the electric circuit and positioned so as to minimize possible damage and/or risk of injury		N/A
	Battery compartments with parallel connections are not recommended since a wrongly placed battery will result in charging conditions		N/A

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Equipment designed to be powered by air-depolarized batteries of either the "A" or "P" system shall provide for adequate air access		N/A
	For the "A" system, the battery should preferably be in an upright position during normal operation		N/A
	For "P" system batteries conforming to Figure 9 of IEC 60086-2:−, positive contact should be made on the side of the battery, so that air access is not impeded		N/A
	When the battery compartment cannot be completely isolated from the equipment, it should be positioned so as to minimize possible damage		N/A
	The battery compartment should be clearly and permanently marked to show the correct orientation of the batteries		N/A
	Battery compartments should be designed so that a reversed battery will result in no electrical circuit		N/A
	The associated circuitry should not make physical contact with any part of the battery except at the surfaces intended for this purpose		N/A
	Designers are strongly advised to refer to IEC 60086-4 and IEC 60086-5 for comprehensive safety considerations		N/A
B.2.2	Limiting access by children		N/A
	Apparatus should be designed to prevent children from removing the battery by one of the following methods		N/A
	A tool, such as a screwdriver or coin, is required to open the battery compartment; or		N/A
	The battery compartment door/cover requires the application of a minimum of two independent and simultaneous movements of the securing mechanism to open by hand		N/A
	If screws or similar fasteners are used to secure the door/cover providing access to the battery compartment, the fasteners should be captive to ensure that they remain with the door/cover		N/A
	This does not apply to side panel doors on larger devices which are necessary for the functioning of the equipment and which are not likely to be discarded or left off the equipment		N/A
B.3	**Voltage cut-off**		**N/A**
	The equipment voltage cut-off should not be below the battery manufacturers' recommendation		N/A
ANNEX C	**DESIGNATION SYSTEM (NOMENCLATURE)**	According to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
C.1	**General**		**P**
C.2	**Designation system in use up to October 1990**		**N/A**
C.2.1	General		N/A
C.2.2	Cells		N/A
C.2.3	Electrochemical system		N/A
C.2.4	Batteries		N/A
C.2.5	Modifiers		N/A
C.2.6	Examples		N/A
C.3	**Designation system in use since October 1990**		**P**
C.3.1	General		P
C.3.2	Round batteries		P
C.3.2.1	Round batteries with diameter and height less than 100 mm		P
C.3.2.1.1	General...:		P
C.3.2.1.2	Method for assigning the diameter code		P
C.3.2.1.3	Method for assigning the height code		P
C.3.2.2	Round batteries with diameter and/or height over or equal to 100 mm		N/A
C.3.2.2.1	General...:		N/A
C.3.2.2.2	Method for assigning the diameter code		N/A
C.3.2.2.3	Method for assigning the height code		N/A
C.3.3	Non-round batteries		N/A
C.3.3.1	General		N/A
C.3.3.2	Non-round batteries with dimensions < 100 mm.....:		N/A
C.3.3.3	Non-round batteries with dimensions ≥ 100 mm.....:		N/A
C.3.4	Ambiguity		N/A
ANNEX D	**STANDARD DISCHARGE VOLTAGE US – DEFINITION AND METHOD OF DETERMINATION**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
D.1	**Definition**		**P**
D.2	**Determination**		**P**
D.2.1	General considerations: the C/R-plot		P
D.2.2	Determination of the standard discharge resistor Rs		N/A
D.2.3	Determination of the standard discharge capacity Cs and standard discharge time ts		P
D.3	**Experimental conditions to be observed and test results**		**P**

IEC/EN 60086-1			
Clause	Requirement + Test	Result - Remark	Verdict
ANNEX E	**PREPARATION OF STANDARD METHODS OF MEASURING PERFORMANCE (SMMP) OF CONSUMER GOODS**	Provided in the battery specification, which will be considered during the end battery manufacturers, distributors, users, and equipment designers.	**P**
E.1	**General**	Complied.	**P**
E.2	**Performance characteristics**	Complied.	**P**
E.3	**Criteria for the development of test methods**	Complied.	**P**
ANNEX F	**CALCULATION METHOD FOR THE SPECIFIED VALUE OF MINIMUM AVERAGE DURATION**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**N/A**
ANNEX G	**CODE OF PRACTICE FOR PACKAGING, SHIPMENT, STORAGE, USE AND DISPOSAL OF PRIMARY BATTERIES**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**N/A**
G.1	**General**		**N/A**
G.2	**Packaging**		**N/A**
G.3	**Transport and handling**		**N/A**
G.4	**Storage and stock rotation**		**N/A**
G.5	**Displays at sales points**		**N/A**
G.6	**Selection, use and disposal**		**N/A**
G.6.1	Purchase		N/A
G.6.2	Installation		N/A
G.6.3	Use		N/A
G.6.4	Replacement		N/A
G.6.5	Disposal		N/A

IEC/EN 60086-2			
Clause	Requirement + Test	Result - Remark	Verdict

6	**PHYSICAL AND ELECTRICAL SPECIFICATIONS**		**P**
6.1	**Category 1 batteries**	Not such category.	N/A
6.1.1	General	See below.	N/A
6.1.2	Specifications: LR20, R20P, R20S.........................:	Not such category.	N/A
6.1.3	Specifications: LR14, R14P, R14S.........................:	Not such category.	N/A
6.1.4	Specifications: LR6, FR14505, R6P, R6S..............:	Not such category.	N/A
6.1.5	Specifications: LR03, FR10G445, R03...................:	Not such category.	N/A
6.1.6	Specifications: LR1, R1, LR8D425..........................:	Not such category.	N/A
6.2	**Category 2 batteries**	Not such category.	**N/A**
	Specifications: CR14250, CR15H270, CR17345, CR17450, BR17335..:		N/A
6.3	**Category 3 batteries**	Not such category.	**N/A**
	Specifications: LR9, CR11108..................................:		N/A
6.4	**Category 4 batteries**	Category 4 – Specifications: CR2032.	**P**
6.4.1	General		N/A
6.4.2	Specifications: PR70, PR41, PR48, PR44...............:		N/A
6.4.3	Fit acceptance gauge for PR batteries		N/A
6.4.4	Specifications: LR41, LR55, LR54, LR43, LR44.....:		N/A
6.4.5	Specifications: SR62, SR63, SR65, SR64, SR60, SR67, SR66, SR58, SR68, SR59, SR69, SR41, SR57, SR55, SR48, SR54, SR42, SR43, SR44.....:		N/A
6.4.6	Specifications: CR1025, CR1216, CR1220, CR1616, CR2012, CR1620, CR2016, CR2032, CR2320, CR2032, CR2330, CR2430, CR2354, CR3032, CR2450, BR1225, BR2016, BR2320, BR2325, BR3032...:	CR2032	P
6.5	**Category 5 batteries**	Not such category.	**N/A**
6.5.1	Specifications: 4LR44, 2CR13252, 4SR44.............:		N/A
6.5.2	Specifications: 5AR40...:		N/A
6.6	**Category 6 batteries**	Not such category.	**N/A**
6.6.1	Specifications: 3R12P, 3R12S, 3LR12....................:		N/A
6.6.2	Specifications: 4LR61...:		N/A
6.6.3	Specifications: CR-P2...:		N/A
6.6.4	Specifications: 2CR5...:		N/A
6.6.5	Specifications: 4R25X, 4LR25X...............................:		N/A
6.6.6	Specifications: 4R25Y...:		N/A
6.6.7	Specifications: 4R25-2, 4LR25-2.............................:		N/A
6.6.8	Specifications: 6F22, 6LR61, 6LP3146...................:		N/A

IEC/EN 60086-2			
Clause	Requirement + Test	Result - Remark	Verdict
6.6.9	Configurations: Stud for 6F22, 6LR61, 6LP3146		N/A
6.6.10	Specifications: 6AS4...:		N/A
6.6.11	Specifications: 6AS6...:		N/A
ANNEX A	**TABULATION OF BATTERIES BY APPLICATION**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	Each of the Tables A.1 to A.25 lists all the batteries for which there is a discharge test given in this specification for that application		P
	Within each table the batteries are listed in ascending order of nominal voltage and, within each nominal voltage, in ascending order of volume		P
ANNEX B	**CROSS-REFERENCE INDEX**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	Batteries having the same physical dimensions may belong to a different electrochemical system		P
	In order to allow physically interchangeable batteries from different electrochemical systems to be compared in terms of electrical performance, a cross-reference is given in Tables B.1 to B.6		P
	Batteries are ranked per category and in each category by chemistry and by shape/size		P
	Batteries are always ranked by voltage and in each voltage by volume		P
ANNEX C	**INDEX**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	The index in Table C.1 provides for the relation between a particular battery and its physical dimensions and application/service output test requirements		P
ANNEX D	**COMMON DESIGNATION**	Accroding to IEC 60086-2 clause 6.4.6 Category 4 – Specifications: CR2032.	**P**
	The index in Table D.1 provides a cross-reference for IEC and common designations of batteries for marking purposes		P

6.4.6	**Category 4 – Specifications: CR2032**	
Initial discharge test		**P**
Lot code / date of manufacture	- / yyyy-mm-dd	/
Storage conditions / duration	20 ± 2 °C, (55 ± 20) RH % / <= 60 d	/
Discharge Conditions (°C, % RH)	20 ± 2 °C, (55 + 20 / - 40) % RH	/
Date(s) of performance of tests / period	yyyy-mm-dd / -	/
OCV max. (V)	3.7	3.23

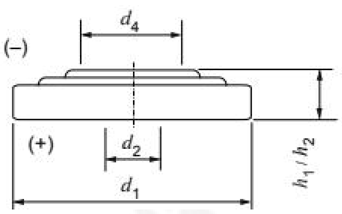
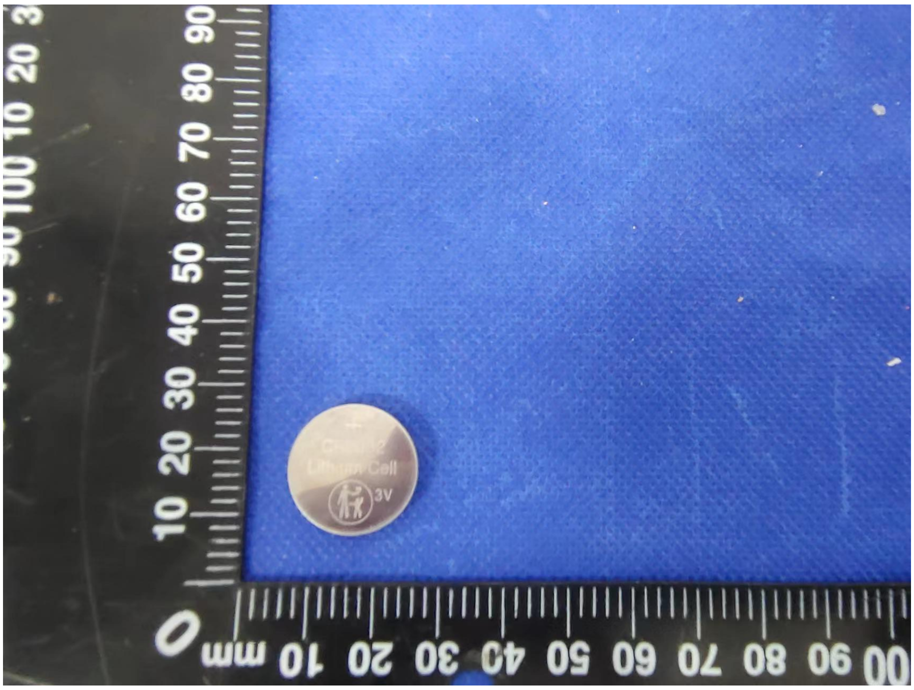


Figure 16 –
Dimensional drawing:
CR1025, CR1216, CR1220, CR1616, CR2012, CR1620, CR2016, CR2032, CR2320, CR2032, CR2330, CR2430, CR2354, CR3032, CR2450, BR1225, BR2016, BR2320, BR2325, BR3032

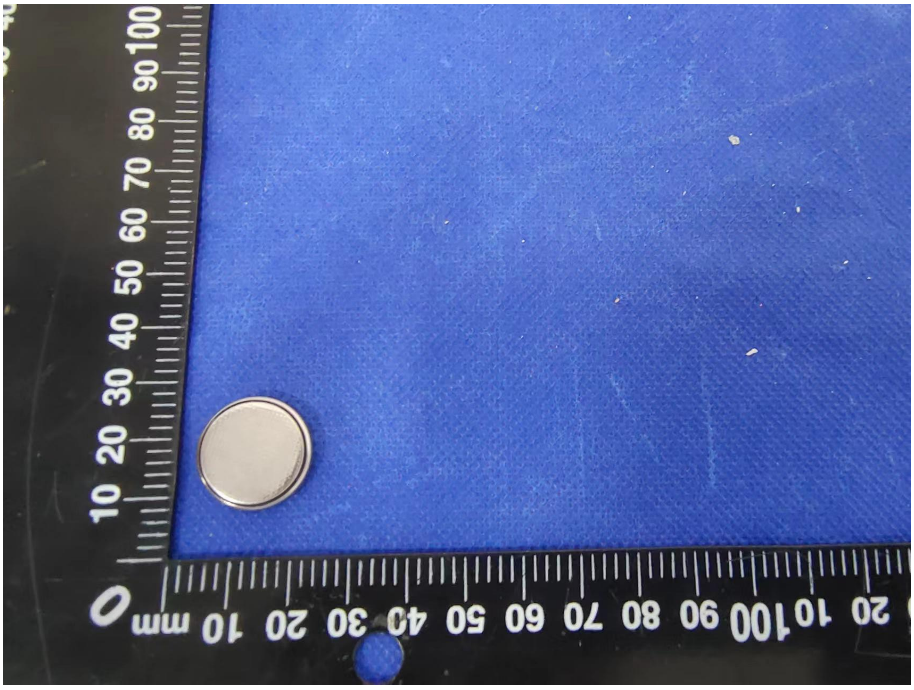
Dimensions (mm)			Initial	After Discharge
				Service output test
h1/h2	max.	3.2	3.16	3.15
	min.	2.9		
d1	max.	16	15.9	15.9
	min.	15.7		
d2	min.	-	-	-
d4	min.	8.0	12.70	12.70
Leakage			No	No

Sample No.	Service output test		Leakage
	Load	15KΩ	
	Daily Period	24h	
	EV (V)	2.0	
	MAD [a] (Initial)	600h	
CR2032 #1	623.5h		No
CR2032 #2	618.6h		No
CR2032 #3	625.1h		No
CR2032 #4	620.7h		No
CR2032 #5	613.3h		No
CR2032 #6	610.5h		No
CR2032 #7	622.4h		No
CR2032 #8	616.3h		No
Average value.	618.8h		No

Notes: a)Standard conditions (see IEC 60086-1:2015, Table 3, Initial discharge test).
Test 8pcs samples and take the average value.



General view-1



General view-2

※※※※※ END OF REPORT ※※※※※